

The Impact of Graduated Backpack Loading on Sagittal Plane Kinematics and Joint Kinetics During Gait

MEEG 482: Clinical Biomechanics

Benjamin Bober, Abigail Denise, Meredith Rhoades, and Connor Spinelli

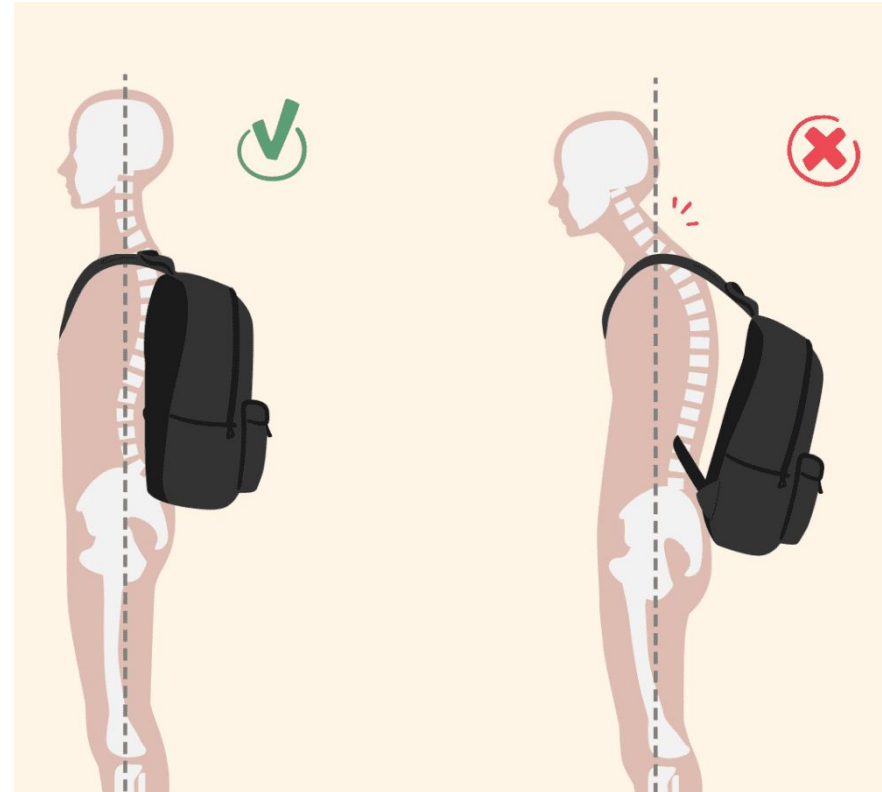


UNIVERSITY OF DELAWARE
ENGINEERING



- Backpack load carriage is common in **students, workers, and military personnel**
- Excessive load has been linked to **altered posture and gait deviations**
- Forward trunk lean and changes in hip and knee angles may **increase** bending moment on the spine, **increase** hip knee and ankle demand, **increased** muscle fatigue
- ★ *Understanding how moderate loads affect gait can guide safer load-carriage recommendations*

- Studies show increased trunk forward lean with loads $\geq 10\text{-}20\%$ **body weight (BW)**
- Hip and knee kinematics can change under load, often increased flexion
- Studies utilize treadmill walking
- Limited research into moderate loads during overground self-selected walking





To quantify how the addition of **external load** **influences the biomechanics** of both the trunk and lower extremities during **level walking**.



1. **Determine** whether moderate backpack loads meaningfully alter sagittal plane gait
2. **Identify** whether a threshold load produces measurable deviations from the baseline and may elevate injury risk
3. **Characterize** how trunk, hip, and knee angles change across the three loading conditions



- Repeated-measures design: each participant completed all three load conditions
- Overground walking at self-selected speed
- 2D sagittal plane video analysis using Kinovea
- Three trials per condition

Subject #	Trial #	Load Conditions
1 ▼	1	0 % BW ▼
	2	0 % BW ▼
	3	0 % BW ▼

Age (yrs)	Sex (M, F, Other)	Body Weight (lbs)
21	M	160
10% body weight	15% body weight	20% body weight
16	24	32
Height (cm)	Leg Length (cm)	Trunk Length (cm)
172	90	40



- Healthy young adults (N = 15)
- Able to walk independently
- Demographics recorded during the testing session

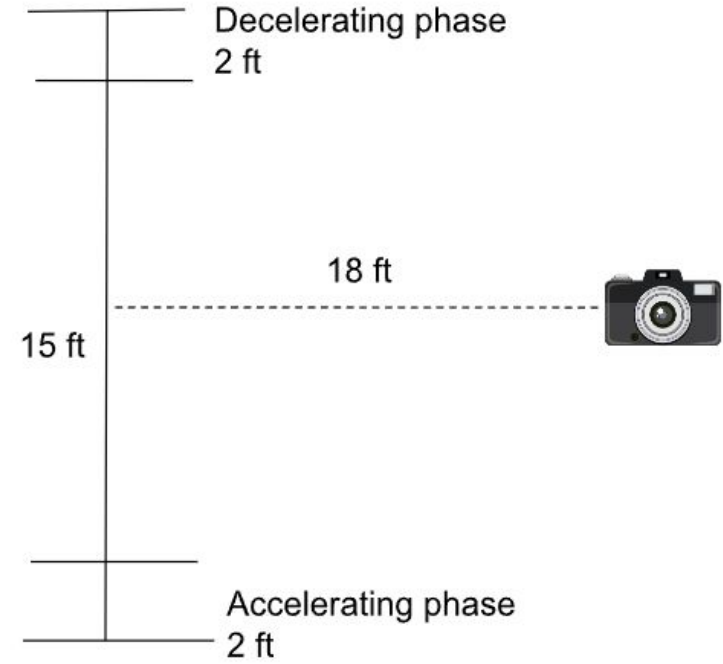
Variable	Mean	Range
Age (years)	21 years	20 - 22 years
Sex (M/F)	M = 5, F = 10	
Height (ft)	5.53 ft	5.00 - 6.00 ft
Leg Length (ft)	2.95 ft	2.79 - 3.28 ft
Body Weight (lbs)	151 lbs	104 - 198 lbs
Walking Speed (ft/s)	4.3 ft/s	3.54 - 5.34 ft/s



- Shoulder Marker
- Hip Marker
- Knee Marker
- Ankle Marker



Total walking
distance
19 ft



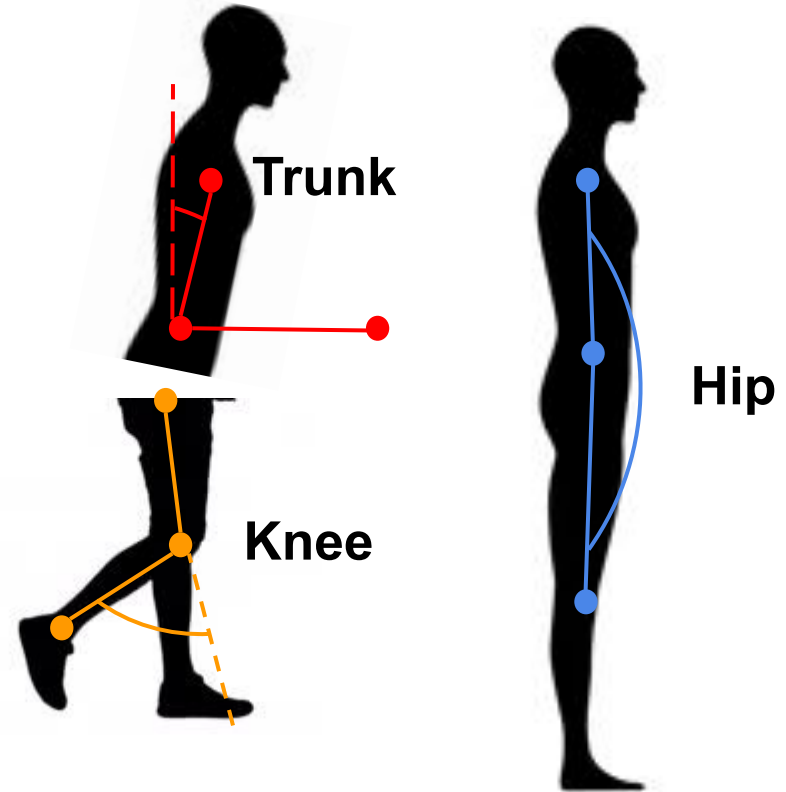
Aerial view



Outcome Measures and Data Analysis

- ★ **Trunk:** Maximum forward lean angle
- ★ **Knee:** Maximum knee angle
- ★ **Hip:** Maximum hip flexion angle

- Means and SDs calculated for each condition
- Repeated-measures ANOVA used to test load effects

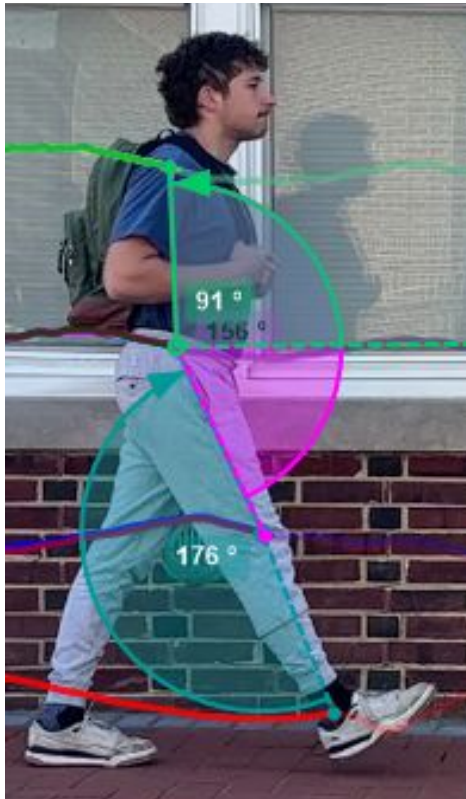




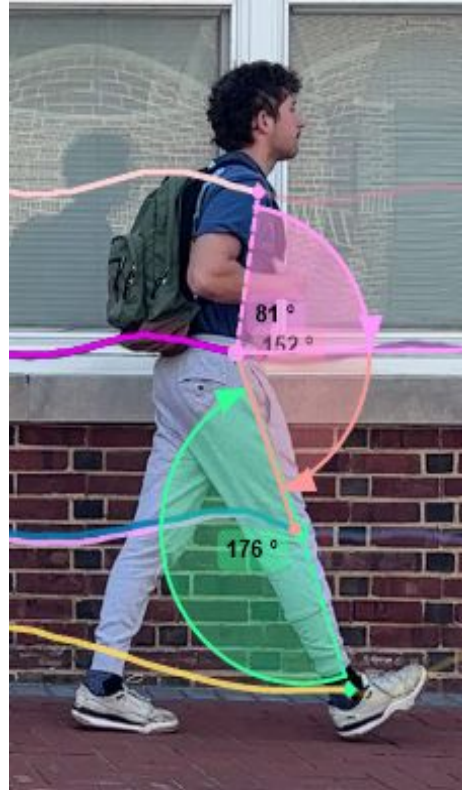
Kinovea analysis with 0% body weight. Slowed to 0.5 speed to see annotations.



Visual Trunk Lean



0% BW



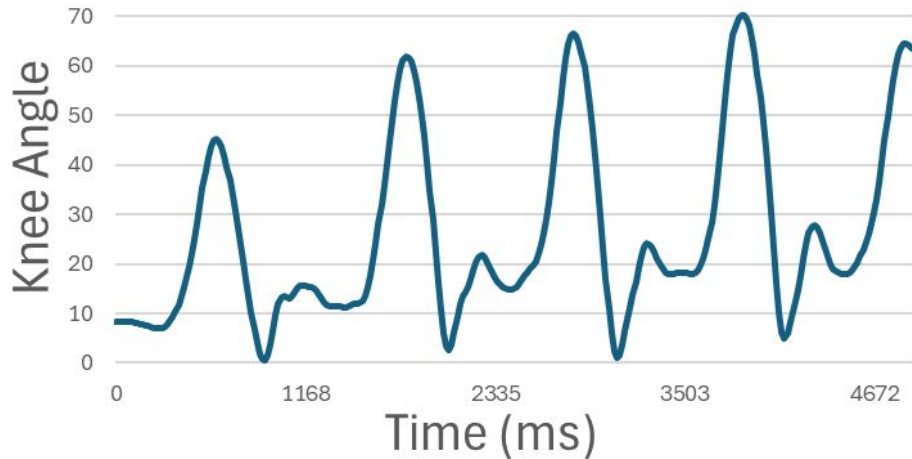
10% BW



20% BW



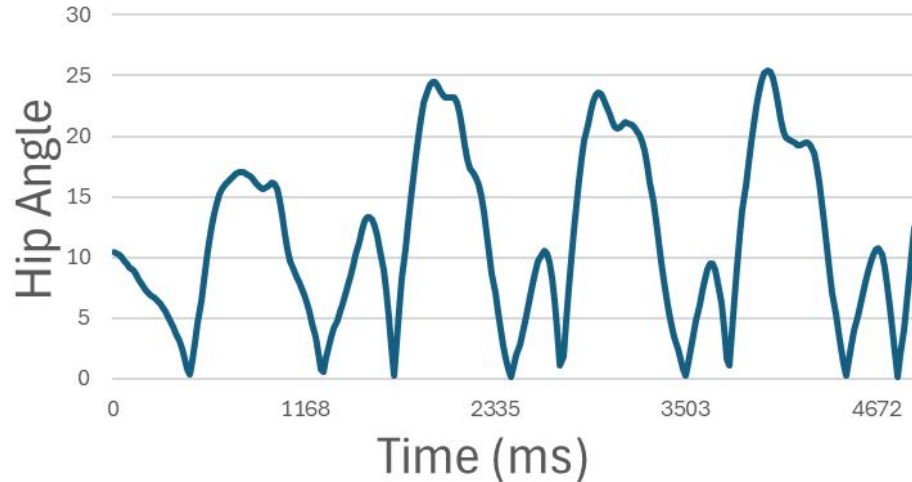
Knee Angle vs. Time (0% BW)



A

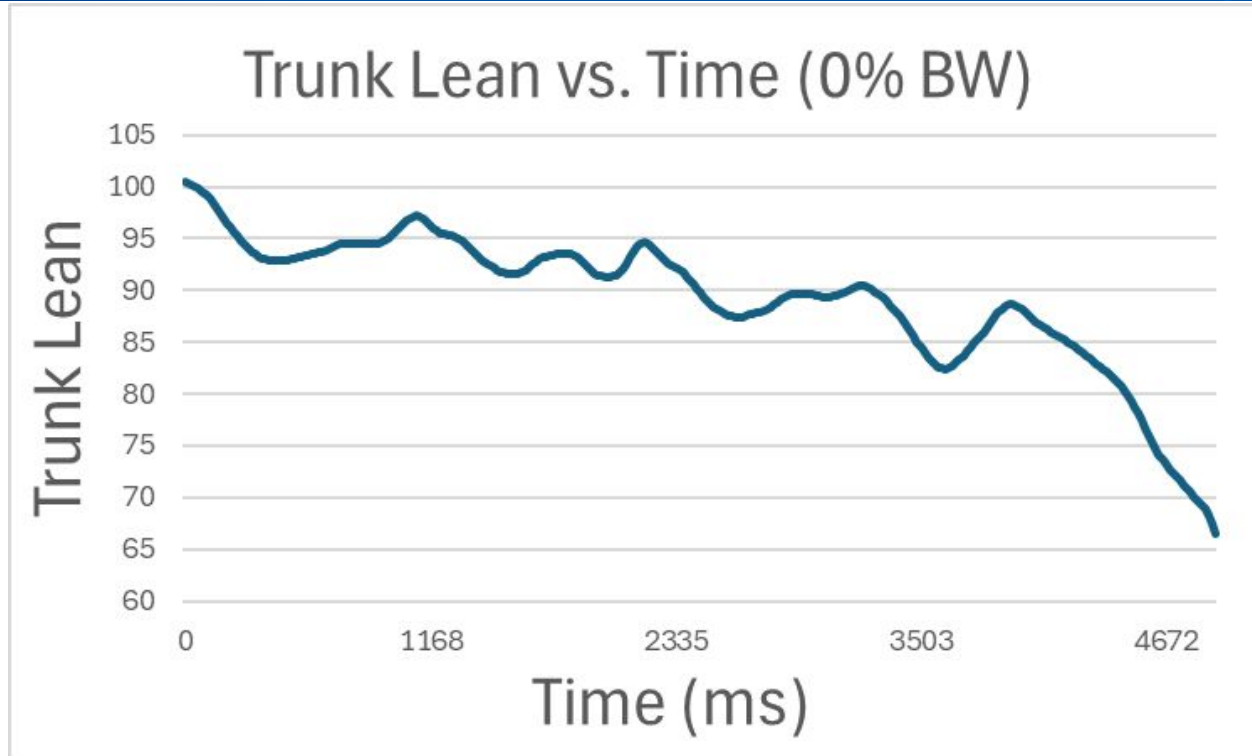
Knee Angle Fluctuations with fluctuations
dependant on gait cycle

Hip Angle vs Time (0% BW)



B

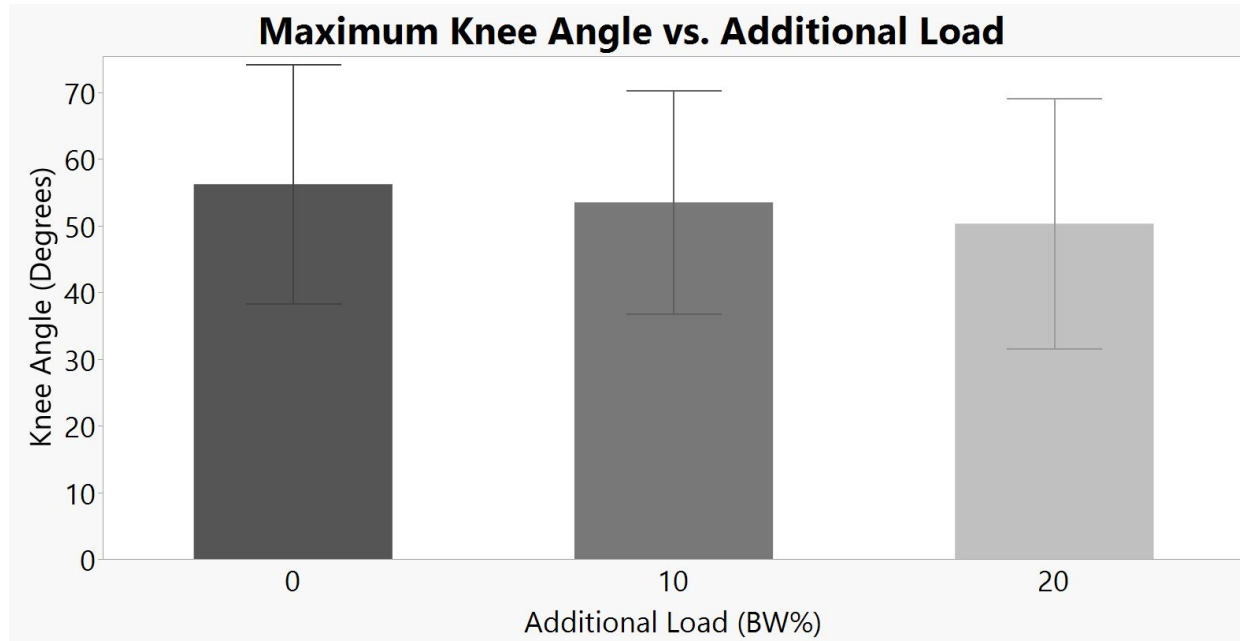
Hip Angle Fluctuations with
fluctuations dependant on gait cycle



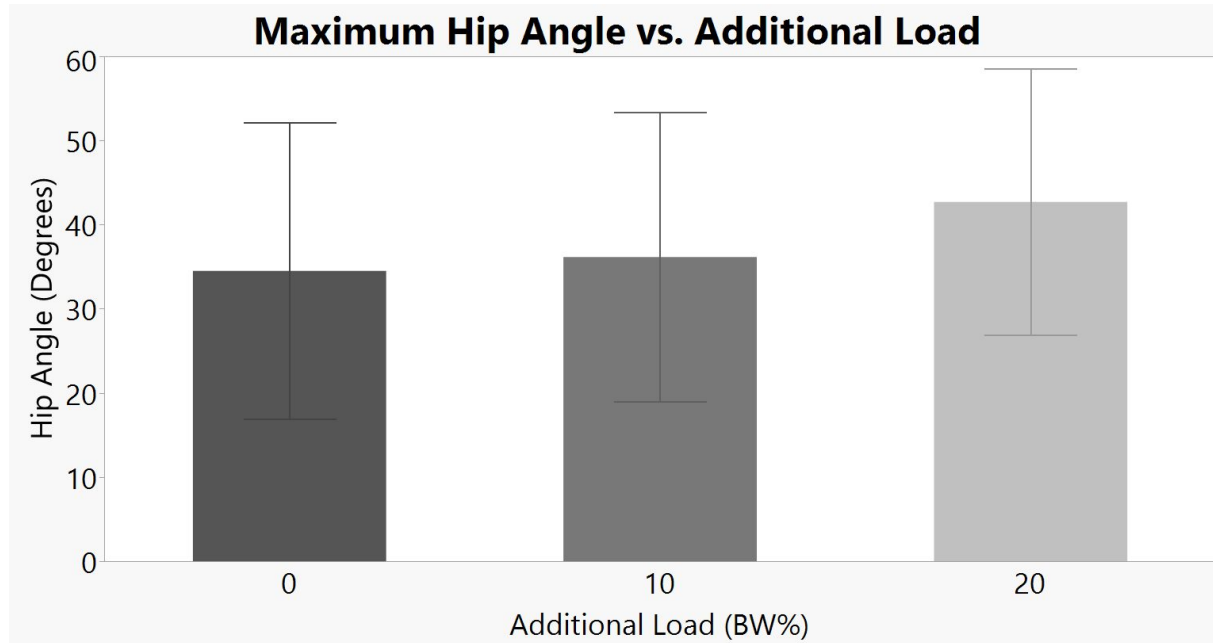
The trunk angle decreases with time proving strenuous loads leads to poor walking mechanics.



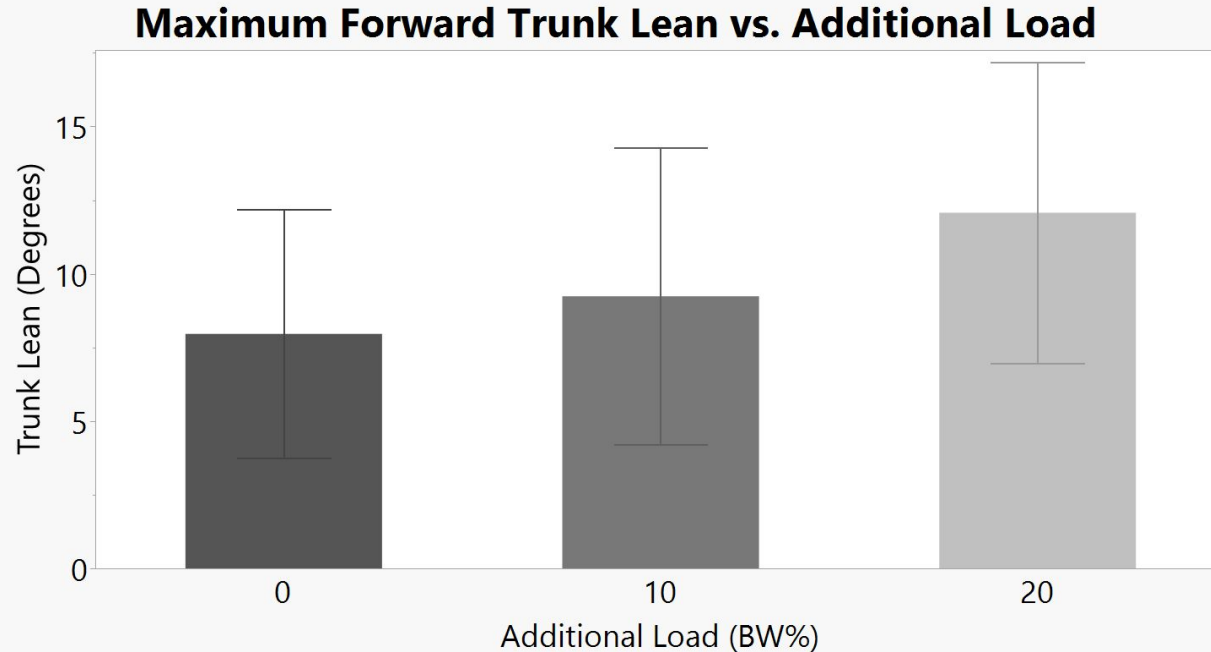
Results: Mean Knee Angle



Mean knee angle decreased slightly with backpack load ($56.21^{\circ} \rightarrow 53.47^{\circ} \rightarrow 50.29^{\circ}$). Data did not show statistically significant results ($p = 0.73$).



Mean hip angle increased with greater backpack load ($34.5^{\circ} \rightarrow 36.17^{\circ} \rightarrow 42.69^{\circ}$). This data did not prove to be statistically significant ($p = 0.78$).



Mean maximum forward trunk lean increased with greater additional backpack load ($7.96^{\circ} \rightarrow 9.24^{\circ} \rightarrow 12.07^{\circ}$) although the effect was not statistically significant ($p = 0.12$).



Previous Studies [3,7]:

Trunk Lean
Increased

Knee Angle
Decreased

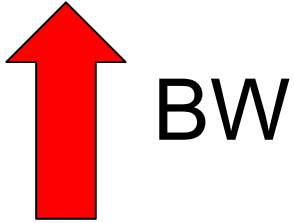
Hip Angle
Increased

Our Study:

Trunk Lean
Increased

Knee Angle
Decreased

Hip Angle
Increased





Study Strengths and Limitations

Strengths:

- Repeated-measures design reduces between subject variability
- Realistic walking conditions: overground, self-paced

Limitations:

- Small sample size
- 2D analysis , reduced precision
- Modest load increments, subtle angle changes
- Task-Inappropriate backpack



Consistent Gait mechanics changes with increasing load

No Statistical Significance however, clear numeric trends

High inter-subject variability, need unique solutions

Overground walking provided more realistic results than treadmill studies

Future research should include larger sample size to identify thresholds

Supports importance of monitoring individual posture and gait

Questions?



UNIVERSITY OF DELAWARE
ENGINEERING



1. Spine Health Foundation. (2024.). *Backpacks and back pain in children* [Photograph]. Spine Health Foundation. <https://spinehealth.org/article/backpacks-back-pain-children/>
2. [1] C. Suri, I. Shojaei, and B. Bazrgari, "Effects of School Backpacks on Spine Biomechanics During Daily Activities: A Narrative Review of Literature," *Hum Factors*, vol. 62, no. 6, pp. 909–918, Sep. 2020, doi: [10.1177/0018720819858792](https://doi.org/10.1177/0018720819858792).
3. [2] L. Ren, R. K. Jones, and D. Howard, "Dynamic analysis of load carriage biomechanics during level walking," *Journal of Biomechanics*, vol. 38, no. 4, pp. 853–863, Apr. 2005, doi: [10.1016/j.jbiomech.2004.04.030](https://doi.org/10.1016/j.jbiomech.2004.04.030).
4. [3] M. Genitrini, F. Dotti, E. Bianca, and A. Ferri, "Impact of backpacks on ergonomics: Biomechanical and physiological effects: A narrative review," *International journal of environmental research and public health*, <https://pmc.ncbi.nlm.nih.gov/articles/PMC9180465/> (accessed Apr. 6, 2026).
5. [4] Dynamic analysis of load carriage biomechanics during level walking - sciencedirect, <https://www.sciencedirect.com/science/article/abs/pii/S0021929004002374> (accessed Apr. 6, 2026).
6. [5] The effect of evenly distributed load carrying on lower body gait dynamics for normal weight and overweight subjects - sciencedirect, <https://www.sciencedirect.com/science/article/abs/pii/S096663621000113X> (accessed Apr. 6, 2026).
7. [6] K. Oberhofer et al., "The influence of backpack weight and hip belt tension on movement and loading in the pelvis and lower limbs during walking," *Applied bionics and biomechanics*, <https://pmc.ncbi.nlm.nih.gov/articles/PMC6011123/> (accessed Apr. 6, 2026).
8. [7] (PDF) influence of backpack load on posture and gait biomechanics in school-aged children, https://www.researchgate.net/publication/392455177_INFLUENCE_OF_BACKPACK_LOAD_ON_POSTURE_AND_GAIT_BIOMECHANICS_IN_SCHOOL-AGED_CHILDREN (accessed Apr. 6, 2026).
9. [8] J. X. Li, Y. Hong, and P. D. Robinson, "The effect of load carriage on movement kinematics and respiratory parameters in children during walking - european journal of applied physiology," *SpringerLink*, <https://link.springer.com/article/10.1007/s00421-003-0848-9> (accessed Apr. 6, 2026).
10. [9] Chow DH;Kwok ML;Au-Yang AC;Holmes AD;Cheng JC;Yao FY;Wong MS;, "The effect of backpack load on the gait of normal adolescent girls," *Ergonomics*, <https://pubmed.ncbi.nlm.nih.gov/16087499/> (accessed Apr. 6, 2026).
11. [10] M. D. Drew, S. M. Krammer, and T. N. Brown, "Effects of Prolonged Walking with Body Borne Load on Knee Adduction Biomechanics," *Gait Posture*, vol. 84, pp. 192–197, Feb. 2021, doi: [10.1016/j.gaitpost.2020.12.004](https://doi.org/10.1016/j.gaitpost.2020.12.004).
12. [11] H.-H. Wang *et al.*, "Effect of Load Carriage Lifestyle on Kinematics and Kinetics of Gait," *Appl Bionics Biomech*, vol. 2023, p. 8022635, Feb. 2023, doi: [10.1155/2023/8022635](https://doi.org/10.1155/2023/8022635).

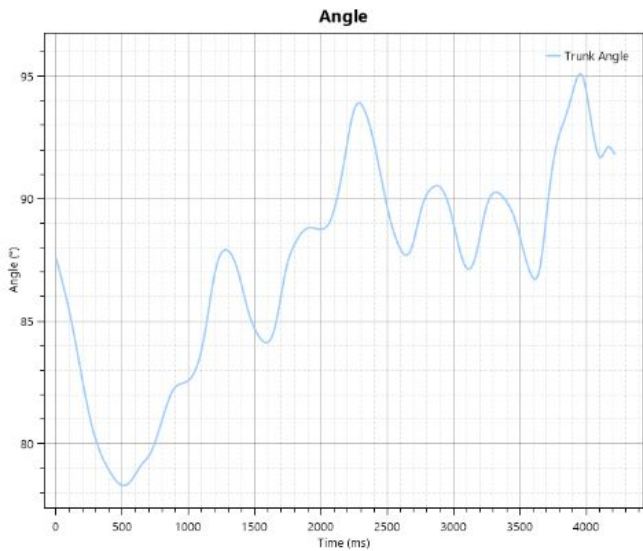
Appendix: Kinovea Trunk Lean Data for 3 subjects



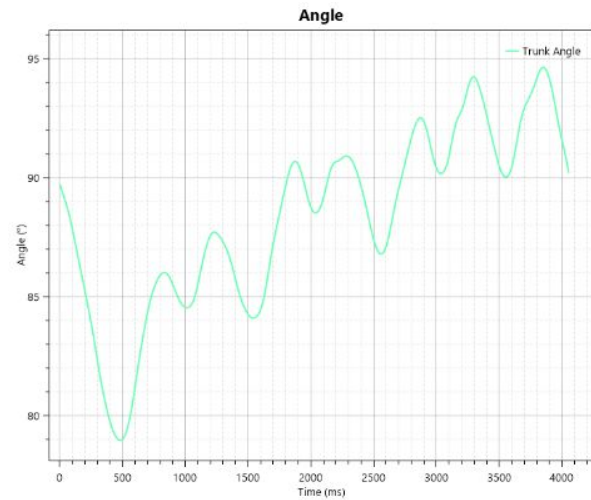
UNIVERSITY OF DELAWARE
ENGINEERING



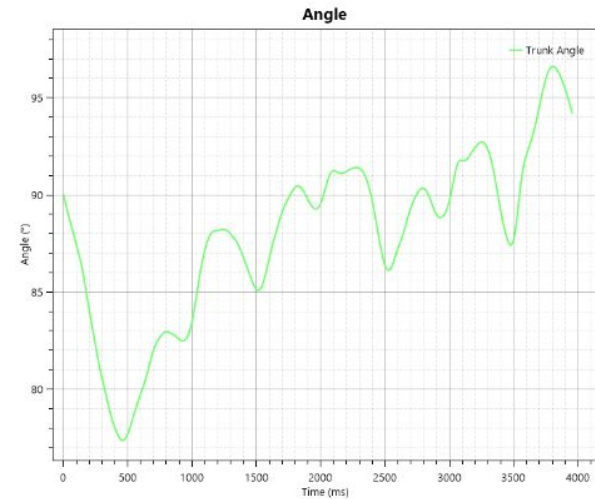
Participant 5 Data



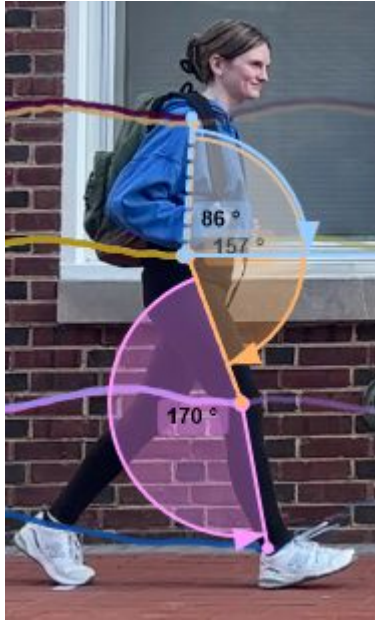
0%



10%



20%



0%



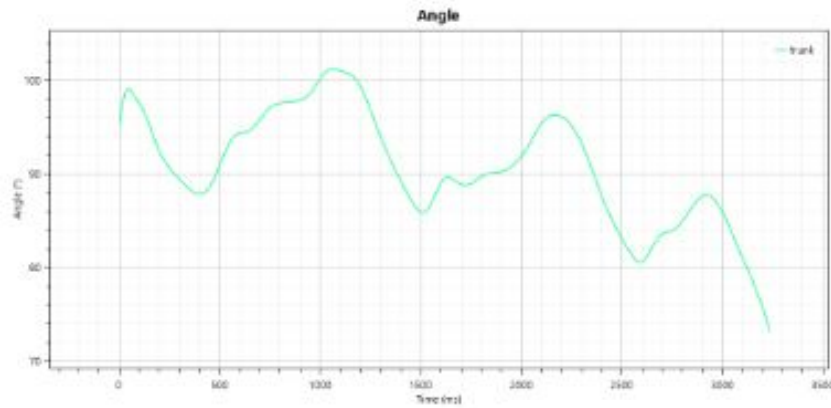
10%



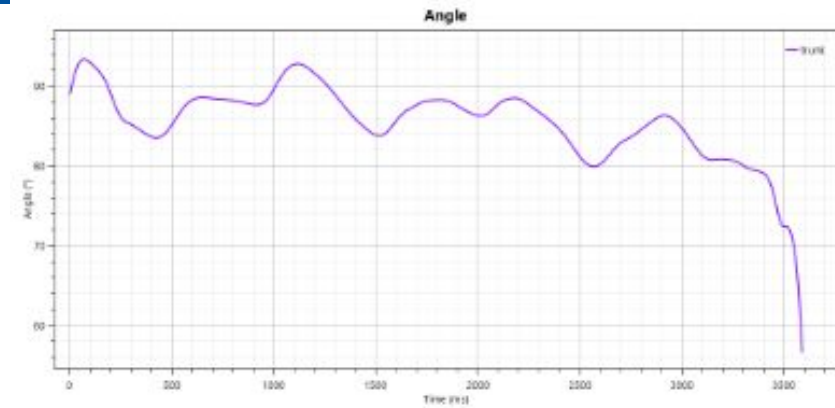
20%



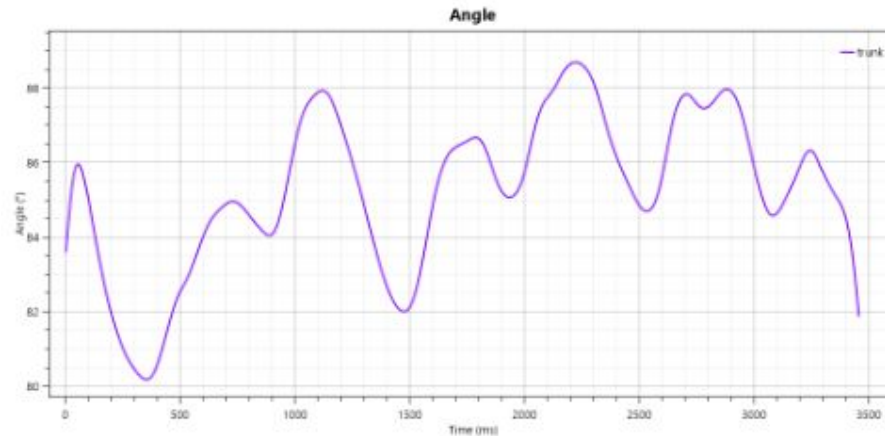
Participant 8 Data



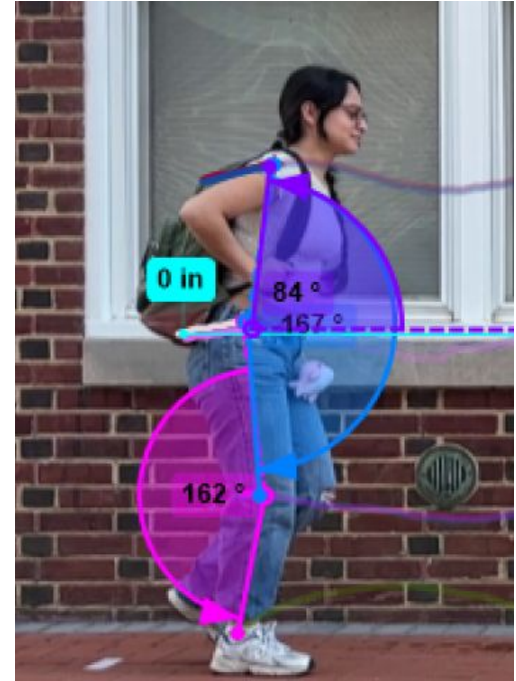
0%



10%



20%

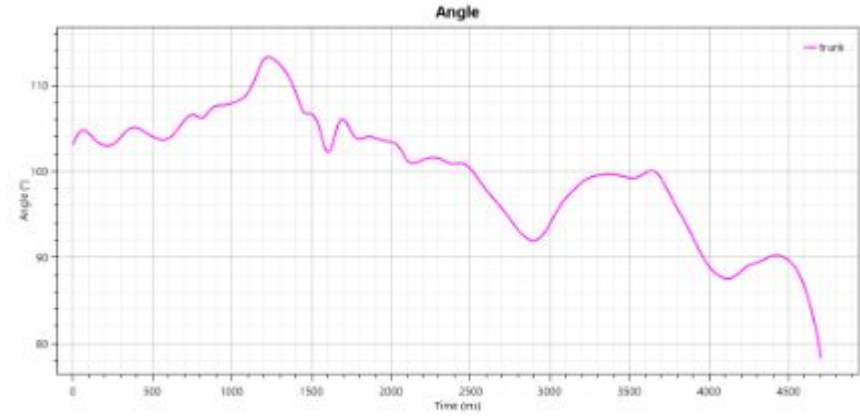




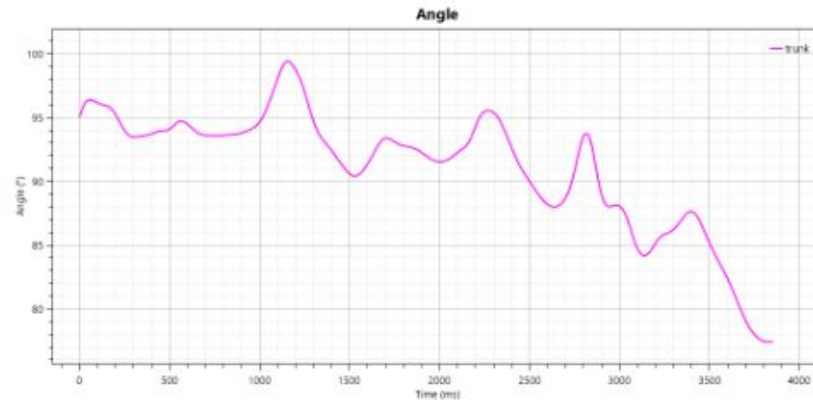
Participant 7 Data



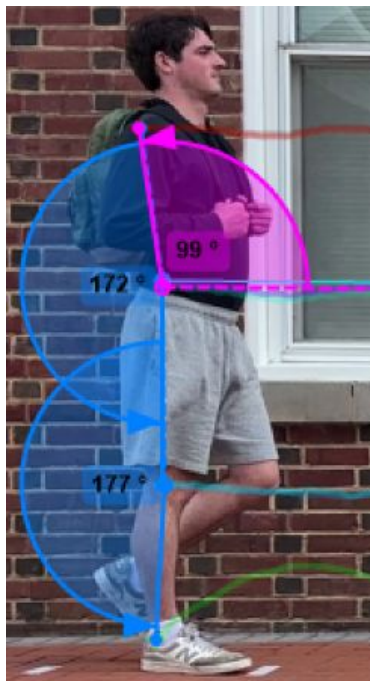
0%



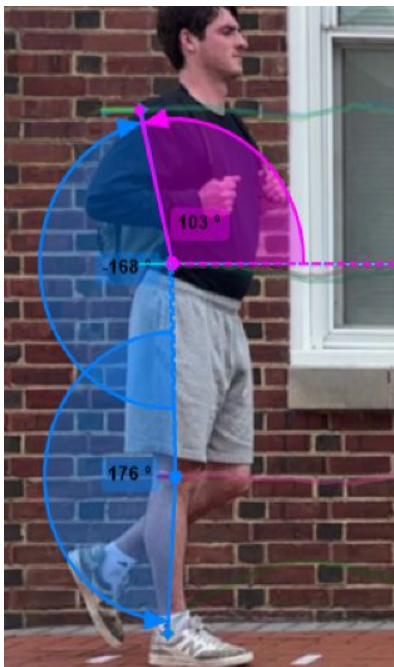
10%



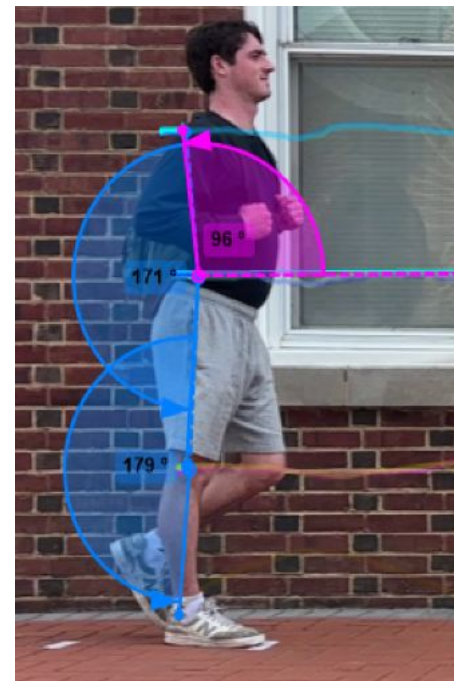
20%



0%



10%



20%